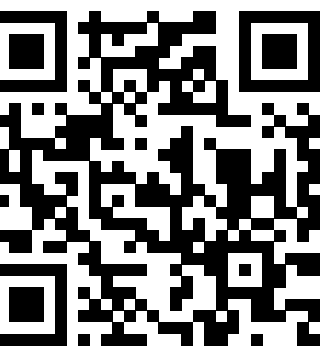
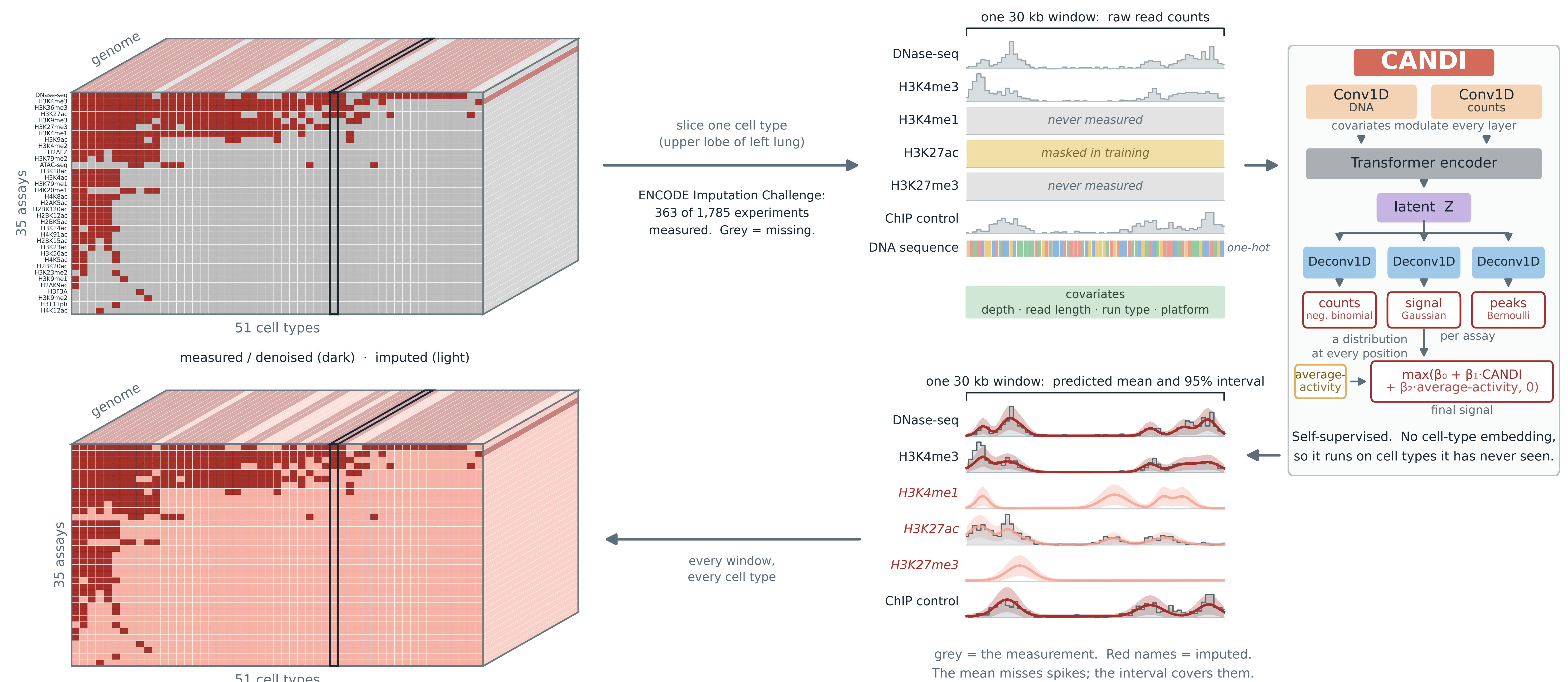


**The problem**

Epigenomic data form a three-dimensional tensor across assays, cell types and genomic positions, with tens to hundreds of assays and thousands of cell types. Due to the high cost and complexity of profiling every possible assay in all relevant cell and tissue types, this tensor is sparse: most experiments were never performed. Epigenome imputation methods aim to predict these missing experiments [1].

Key findings

- CANDI ranks **1st of 26** on the ENCODE Imputation Challenge.
- CANDI provides calibrated uncertainty estimates.
- CANDI's imputed signals and latent space predict gene expression.

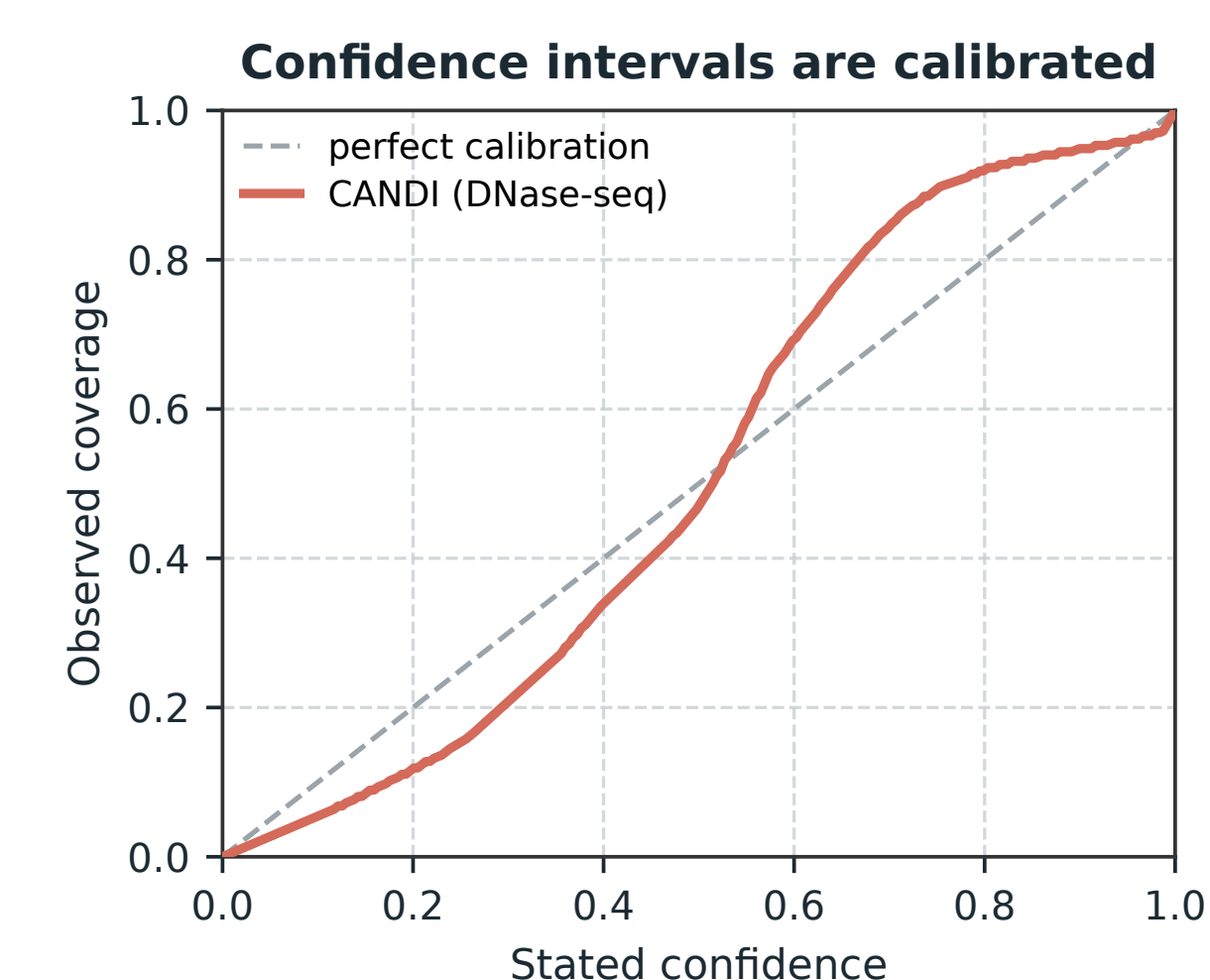
CANDI schematic workflow

[1] Schreiber J, Boix C, *et al.* The ENCODE Imputation Challenge: a critical assessment of methods for cross-cell type imputation of epigenomic profiles. *Genome Biol* 24:79 (2023).

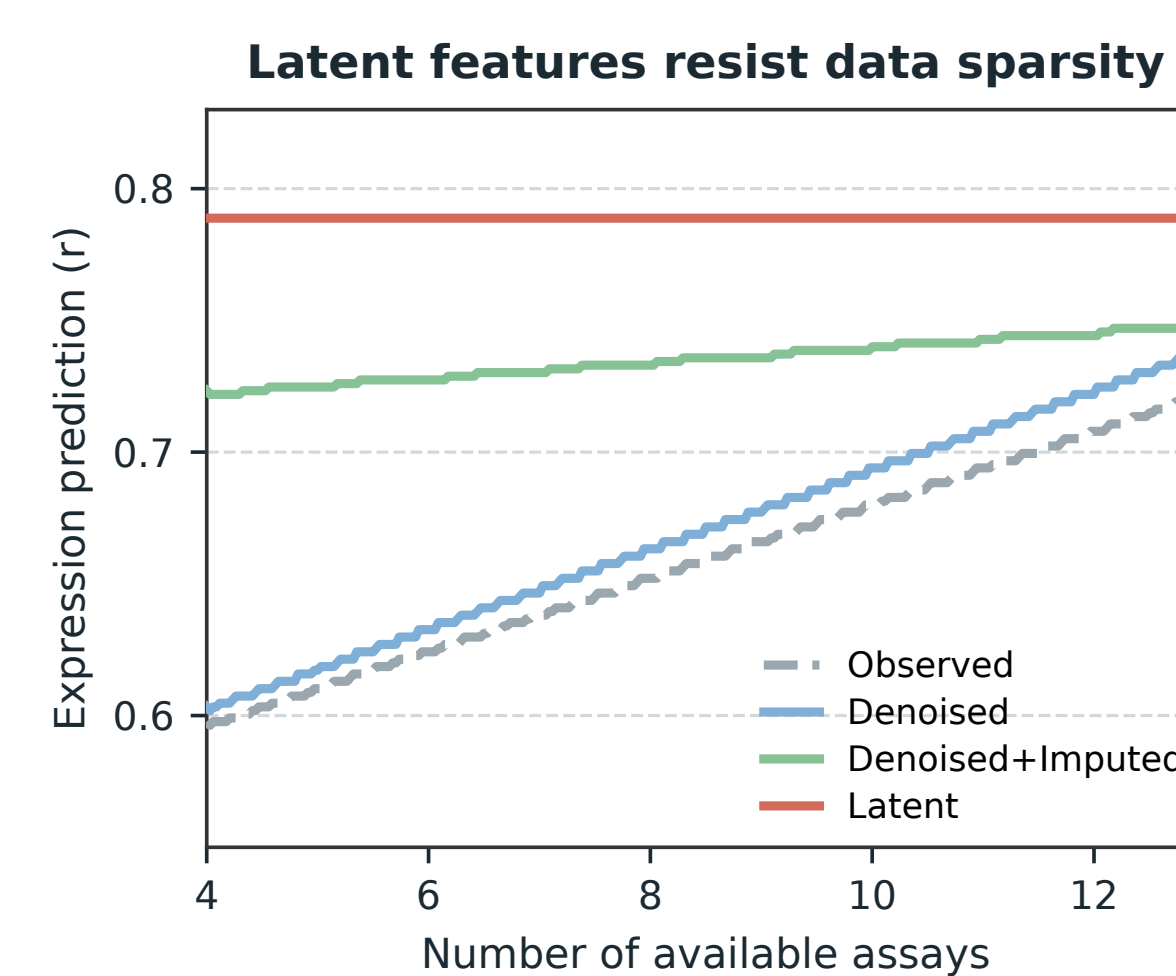
[2] Foroozandeh M, Diab AR, Libbrecht M. CANDI: self-supervised, confidence-aware denoising imputation of genomic data. *bioRxiv* 2025.01.23.634626 (2025).

Panels C–D: blind-test scoring from the Libbrecht lab's EIC reproduction; *baseline* is the challenge's average-activity track. The correction was chosen after the challenge results were public; none of it is fitted on blind-test data.

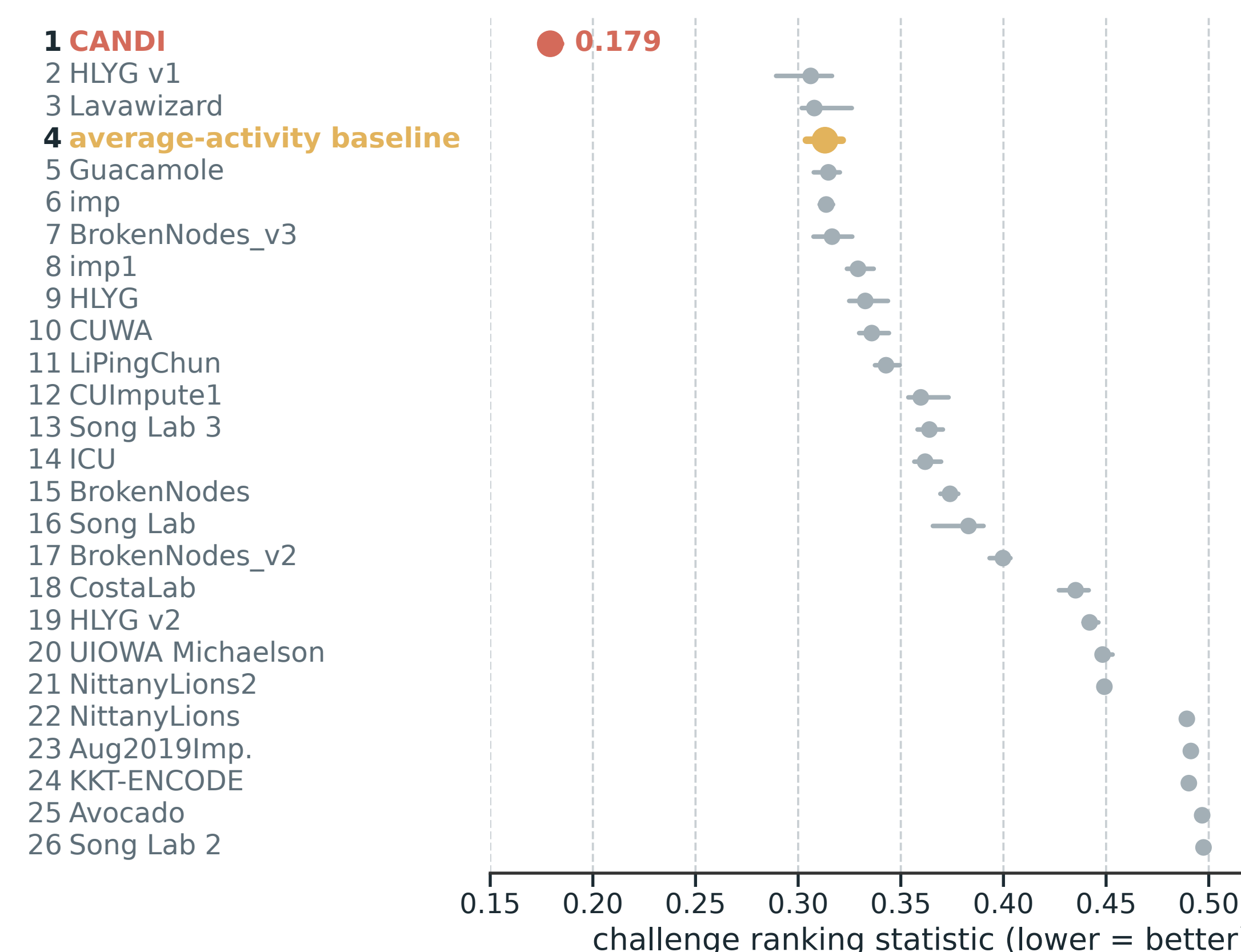
Top: for each 30 kb locus, CANDI takes observed read counts with the ChIP-seq control, four covariates per assay and the DNA sequence; grey assays were never measured, yellow ones are masked. **Bottom:** each output is a probability distribution, here a mean and 95% interval per assay; window by window, they fill every missing thread of the tensor.

Downstream utility of CANDI

A Confidence calibration for DNase-seq (DND-41): predicted confidence level against the fraction of observed data within that interval.



B Gene expression prediction as input assays are removed. Denoised+Imputed and Latent settings are robust to input sparsity.

Fixing the train-test experimental covariate shift in the ENCODE Imputation Challenge

C The leaderboard with CANDI added: CANDI is first at 0.179 (bootstrap range 0.177–0.184), the next entry 0.306.



D Eight of the nine measures relative to the baseline, median over 51 experiments. CANDI is first on three and top 8 on the rest.